

Methodology

Pipeline

Ground truth = Dorado SUP basecaller → minimap2 (RawBench protocol).

RawHash2 invoked with -x sensitive (and --r10 for R10.4.x), Uncalled4 pore model.

Eval = best-mapping per read, 0.1 overlap threshold (TP/FP/FN → F1).

Read-ID-mismatch fix on D3/D4: re-extract reads from FAST5 internal fastq → minimap2.

Segmenters (9, 5 algorithmic families)

- t-test family: default (ONT dual-window), scrappie
- DP-CPD: pelt (BIC penalty), binseg (per-segment penalty fix)
- Probabilistic: hmm (Viterbi+k-means), bocd (Adams & MacKay 2007)
- Sliding-window: window (z-score), mad (median two-sample)
- Edge: gradient (1st derivative)

Top-K mechanism (key engineering)

Replaces threshold-based detection with rank-based selection: pick the $K=n/9$ highest-scoring positions (where 9 = R10 samples-per-k-mer). Greedy NMS with $\text{min_size}=4$ separation. K is derived from pore physics, not from data → not overfit.

Boost on gradient/mad/window: $F1 \approx 0 \rightarrow 0.1-0.5$ (RB_ecoli).

Per-segmenter index

Each segmenter builds its OWN RawHash2 index by simulating reference signal + segmenting it with the same method used at query time. Without per-segmenter index, non-default segmenters give $F1 \approx 0$ due to feature-distribution mismatch.

Cross-validation

3-fold read-level CV on D1/D2/D4/D6/RB_ecoli/RB_dmel for HMM (12 configs:

STATES $\in \{3..8\} \times$ STAY $\in \{0.93, 0.96\}$) and BOCD (6 configs: EXP_LEN $\in \{50..600\}$).

Train-test gap nearly 0 across all configs → no overfit. Best per dataset

selected by mean F1_test (not F1_train).

Datasets in scope (D7 excluded — FAIL reads)

Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh)	R10.4.1	12,000	3.0 GB	7.10 GB

Datasets — extended characterization

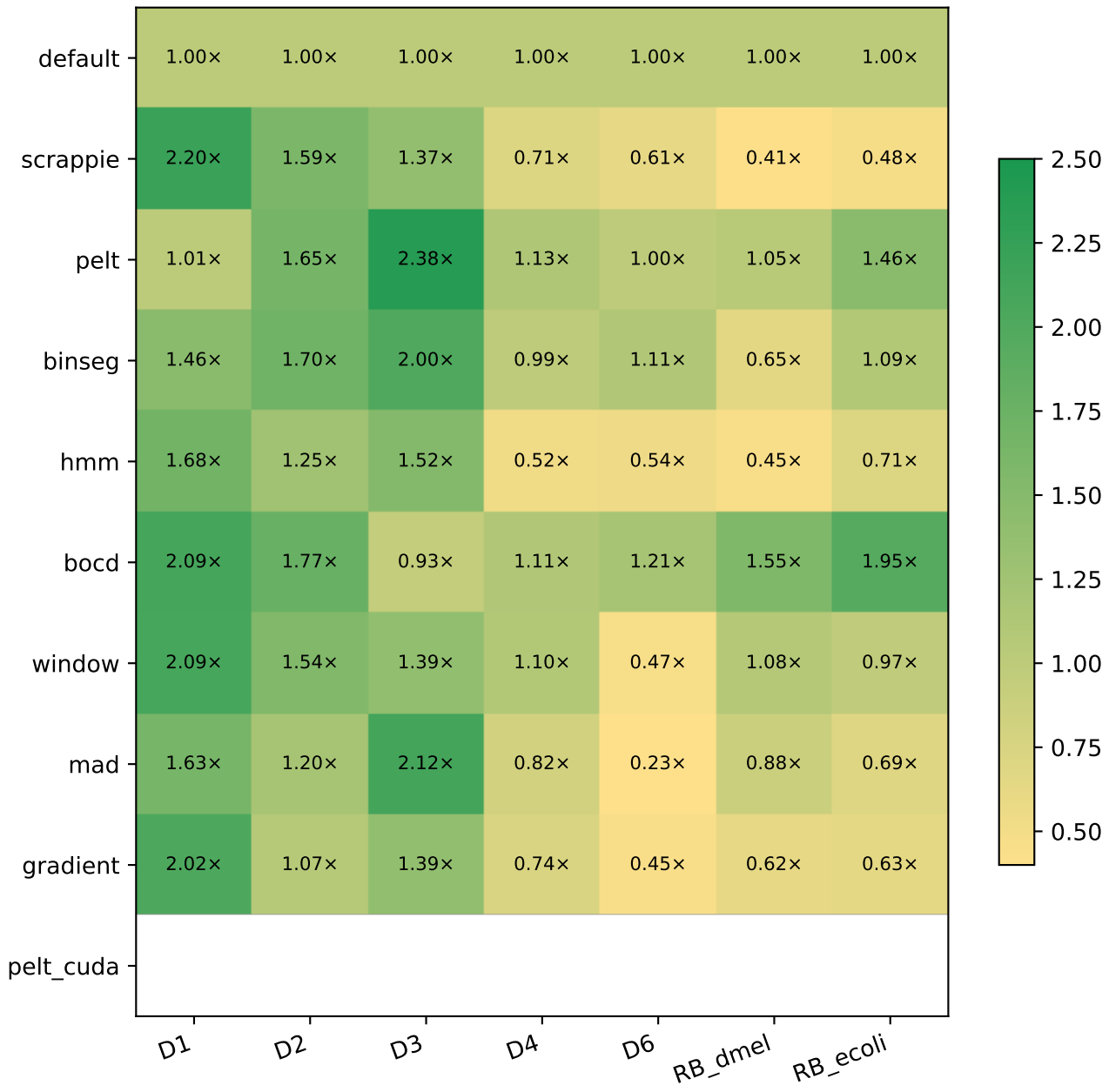
DS	Organism	Chem	Reads	Ref bp	GC%	contigs	N50 (Mb)	N%	masked%
D1	SARS-CoV-2	R9.4	1,383	0.03 Mb	38.0	1	0.03	0.00	0.0
D2	E. coli	R9.4	89	5.23 Mb	50.5	1	5.23	0.00	0.0
D3	S. cerevisiae	R9.4	500	12.16 Mb	38.1	17	0.92	0.00	0.0
D4	C. reinhardtii	R9.4	500	111.10 Mb	64.1	53	7.78	3.65	32.4
D6	E. coli	R10.4	294	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_dmel	D. melanogaster	R10.4.1	12,000	143.73 Mb	42.0	1,870	25.29	0.80	20.9
RB_ecoli	E. coli	R10.4.1	12,000	5.23 Mb	50.5	1	5.23	0.00	0.0

Ref bp = total reference length; *N50* = contig N50; *N%* = ambiguous N bases; *masked%* = soft-masked low-complexity / repeats (lowercase in FASTA).

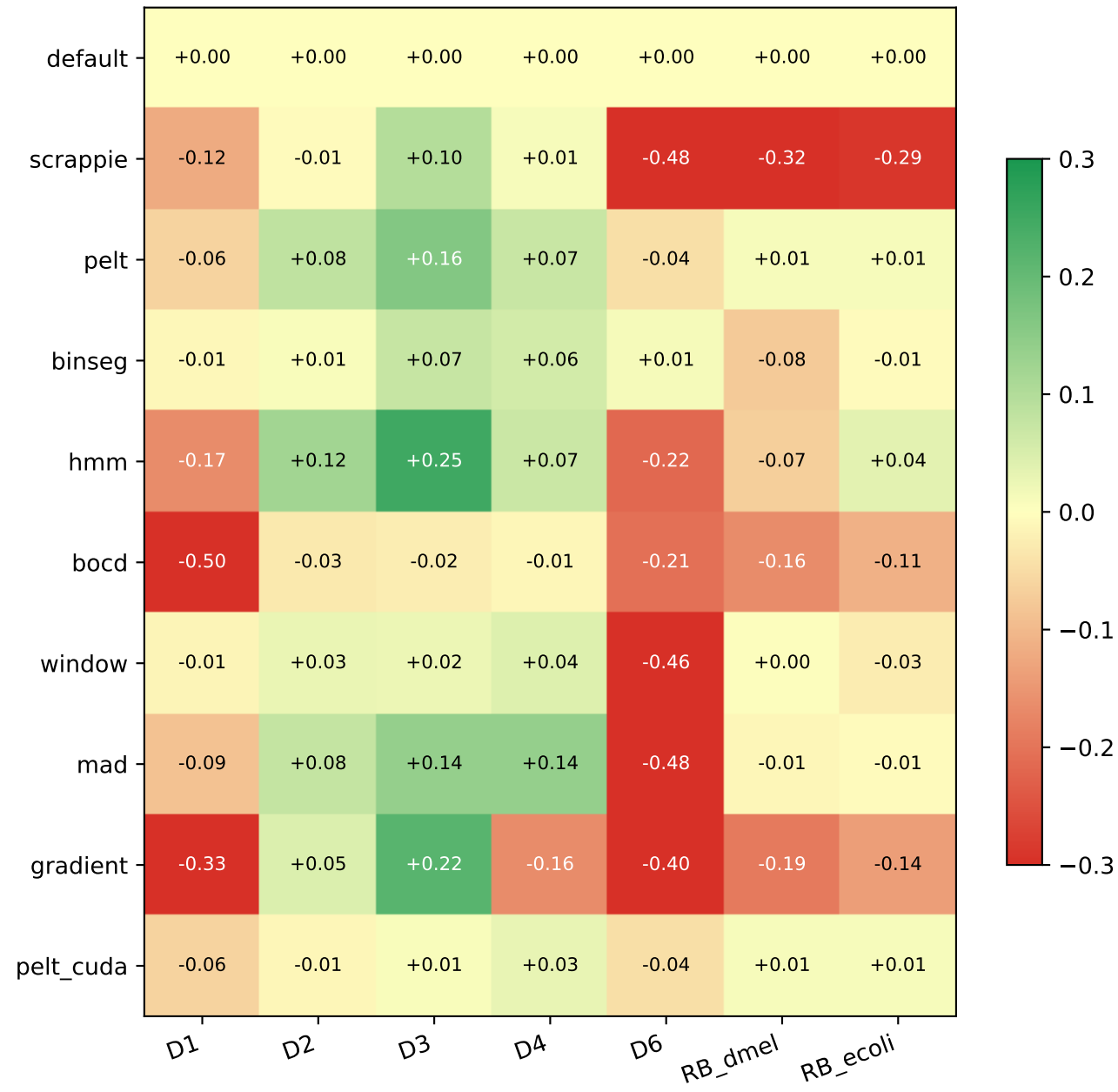
F1 across (segmenter, dataset) — best CV-tuned config used for HMM/BOCD



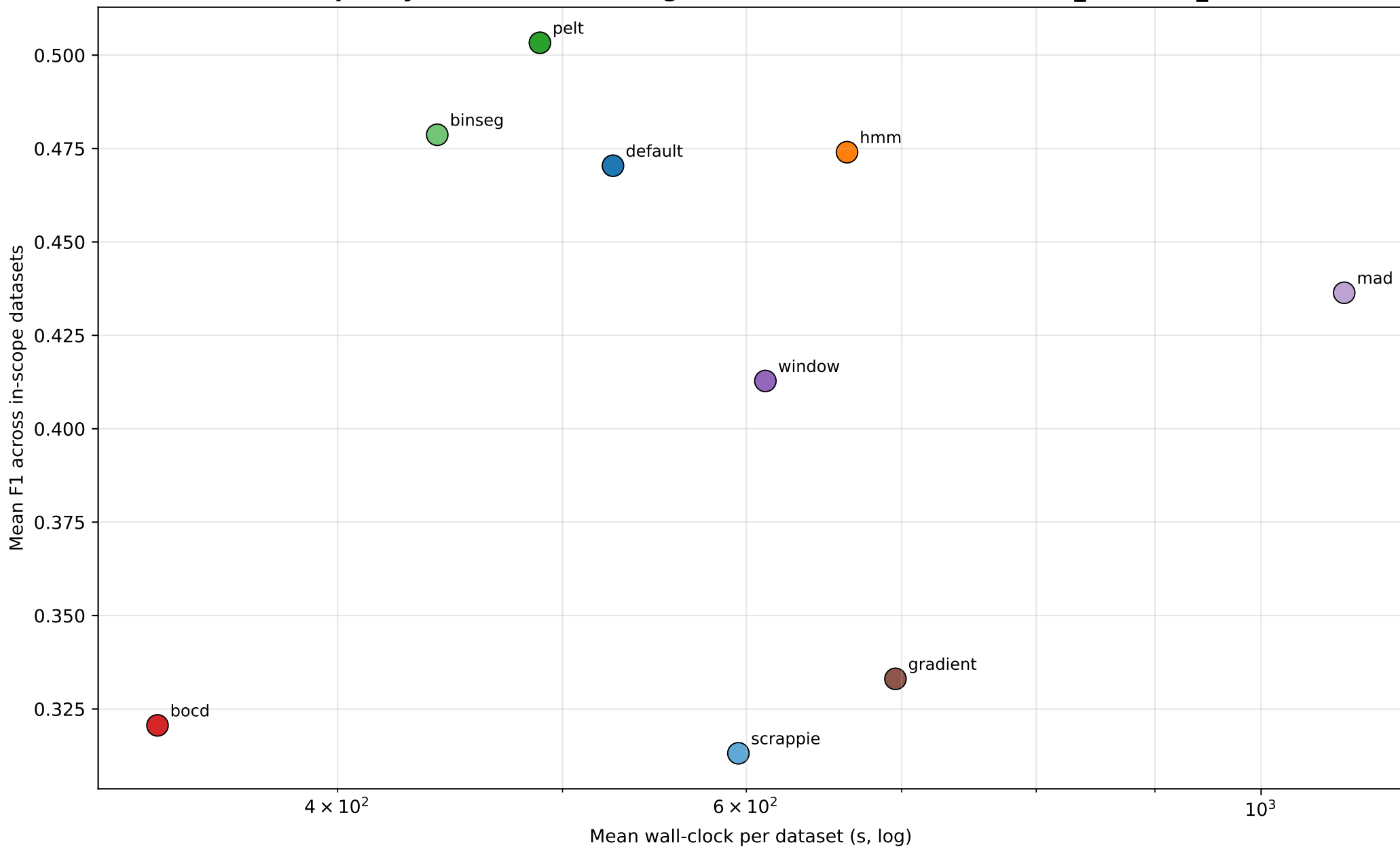
Speedup vs default



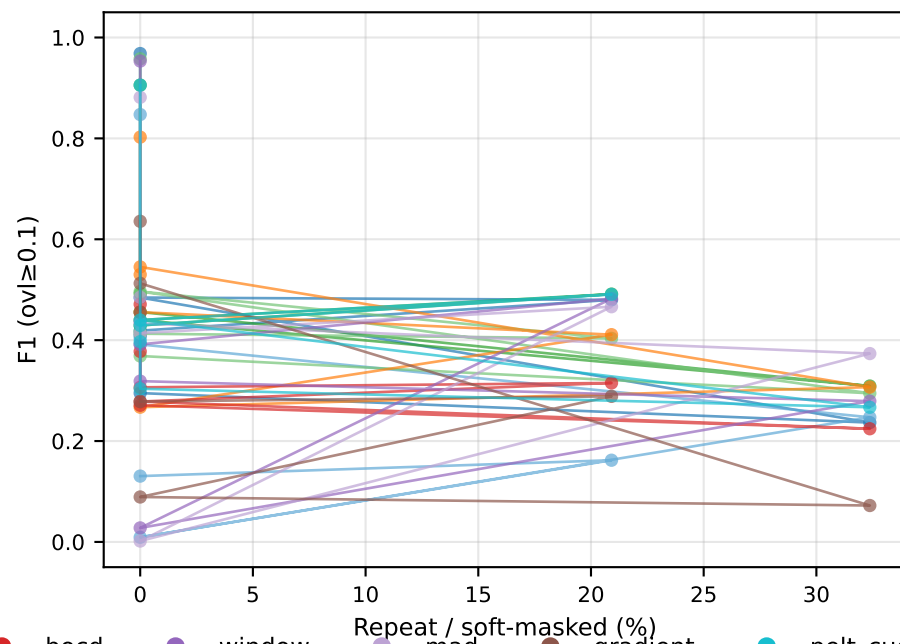
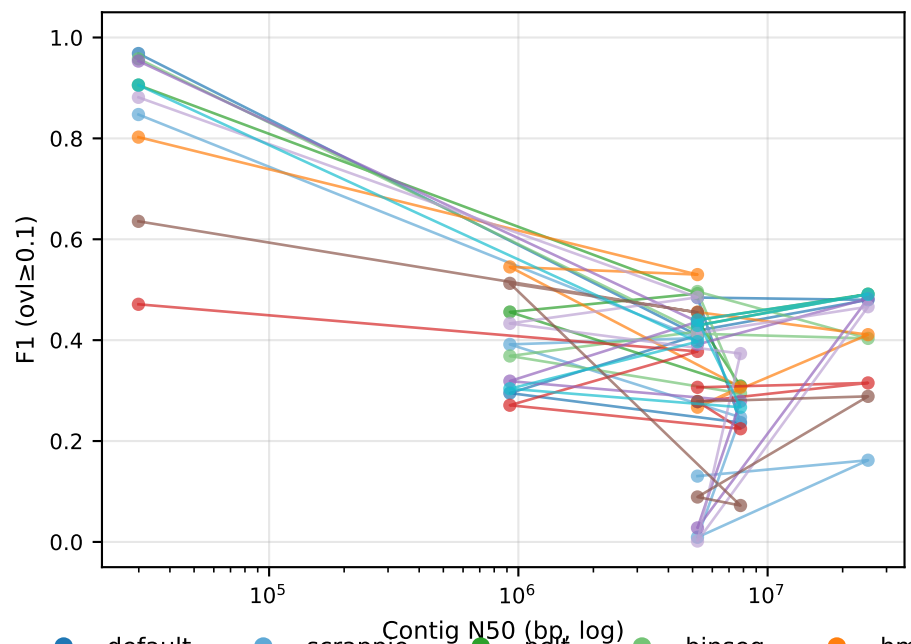
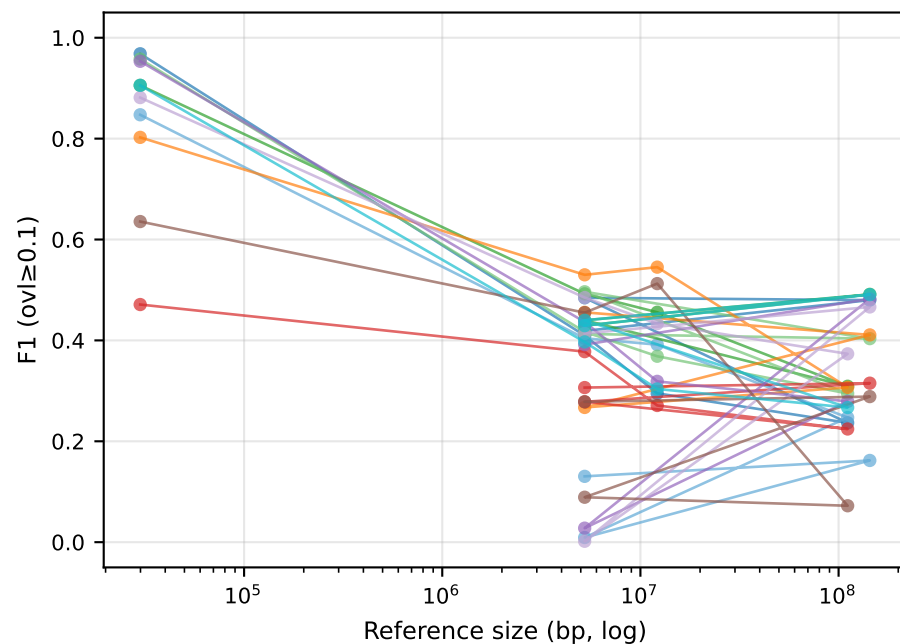
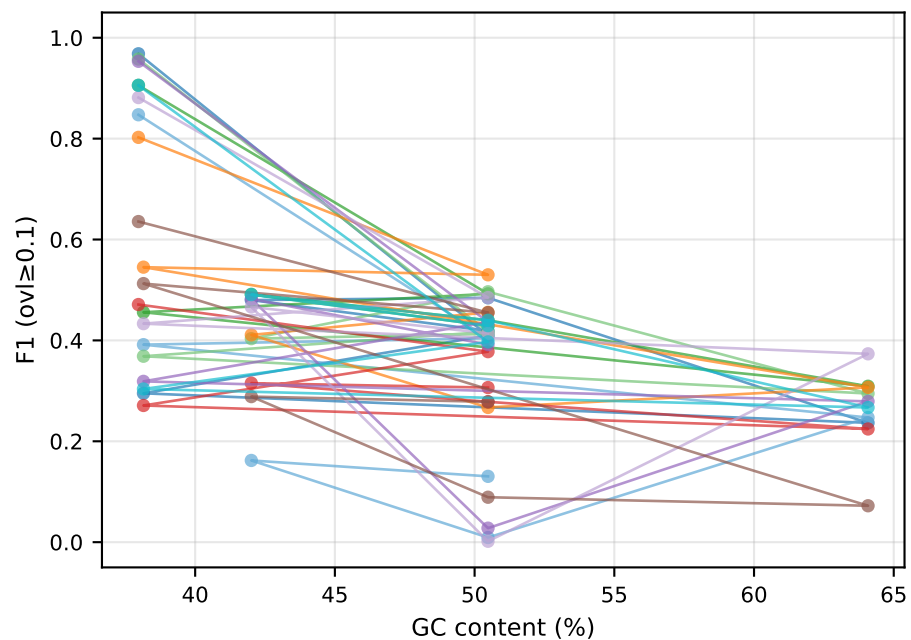
F1 delta vs default



Pareto: quality vs runtime (averaged across D1,D2,D3,D4,D6,RB_dmel,RB_ecoli)



F1 vs reference attributes — per segmenter (Setup A, $ovl \geq 0.1$)

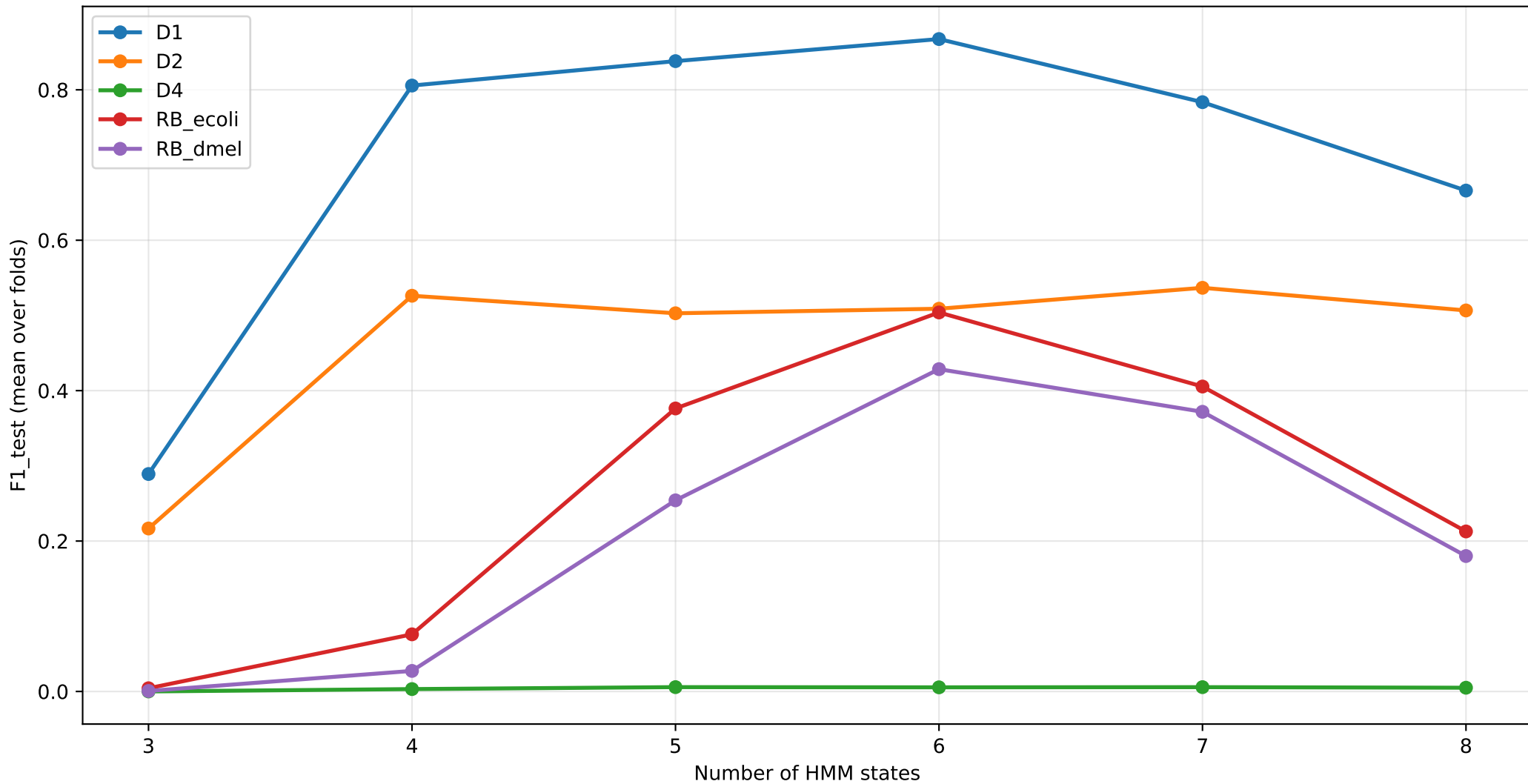


● default
 ● scrappie
 ● pelt
 ● binseg
 ● hmm
 ● bocd
 ● window
 ● mad
 ● gradient
 ● pelt_cuda

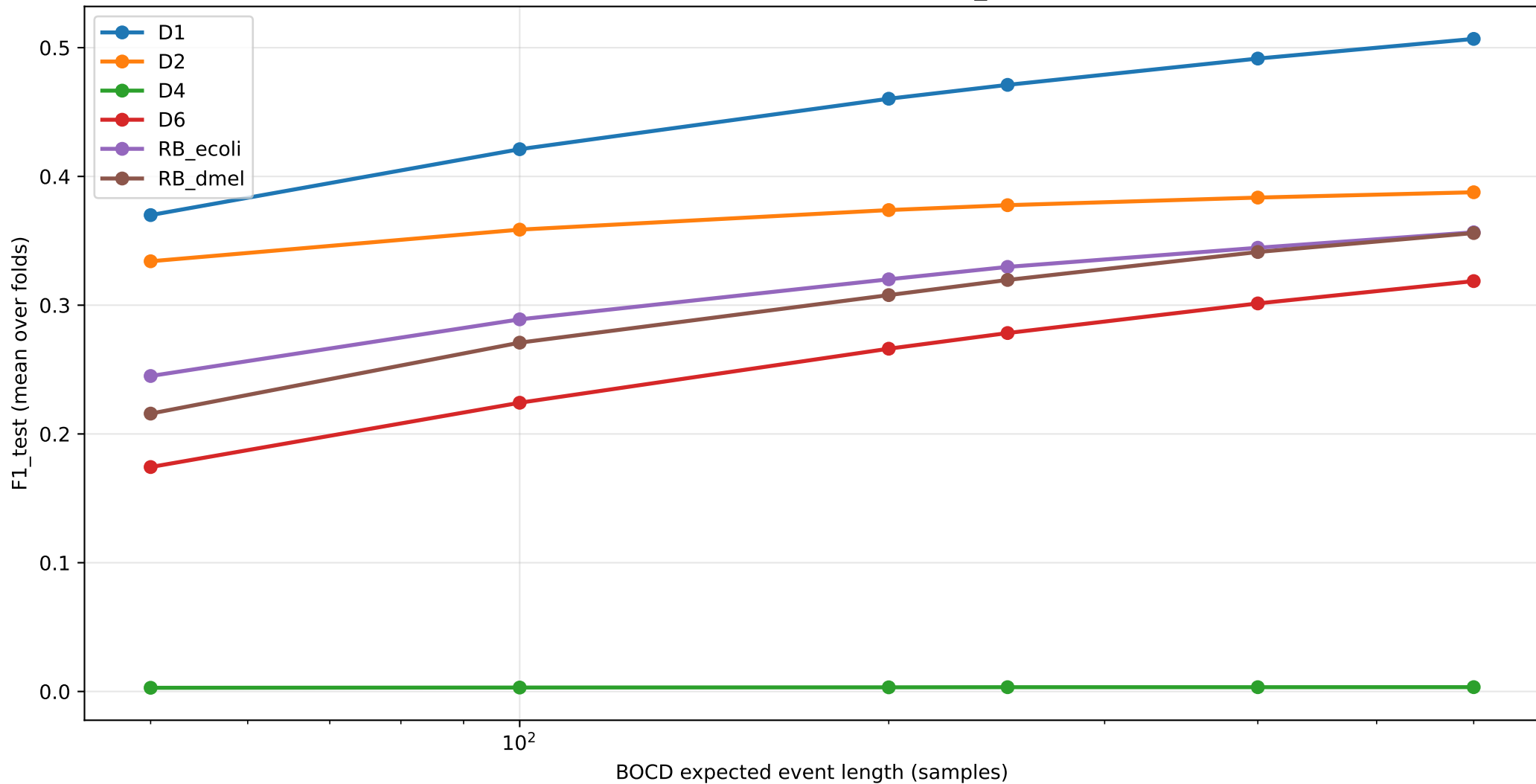
Pearson correlation: F1 (ovl $\geq$ 0.1) vs reference attribute



HMM CV sweep: F1 vs STATES (STAY=0.96)



BOCD CV sweep: F1 vs EXP_LEN



Per-dataset segmenter recommendation (BEST F1 vs FAST-near-default)

Dataset	default (F1 wall)	BEST F1 segmenter	FAST + $\geq 95\%$ F1 segmenter
D1	default (0.968 1359s)	default (0.968 1359s) [1.00×	window (0.953 651s) [2.09×
D2	default (0.410 549s)	hmm (0.530 439s) [1.25×	hmm (0.530 439s) [1.25×
D3	default (0.295 6s)	hmm (0.545 4s) [1.52×	hmm (0.545 4s) [1.52×
D4	default (0.237 13s)	mad (0.373 16s) [0.82×	mad (0.373 16s) [0.82×
D6	default (0.484 1332s)	binseg (0.496 1203s) [1.11×	binseg (0.496 1203s) [1.11×
RB_dmel	default (0.480 366s)	pelt (0.491 350s) [1.05×	window (0.482 339s) [1.08×
RB_ecoli	default (0.419 55s)	hmm (0.456 76s) [0.71×	hmm (0.456 76s) [0.71×

Findings

Top findings

1. HMM with chemistry-aware tuning beats default on bacterial genomes

- D2 R9.4 E. coli: HMM (S=4, stay=0.93) F1=0.530 vs default 0.410 ($\Delta=+0.120$)
- RB_ecoli R10.4.1: HMM (S=6, stay=0.96) F1=0.456 vs default 0.419 ($\Delta=+0.037$)
- D3 yeast R9.4: HMM F1=0.545 vs default 0.295 ($\Delta=+0.250$)
- D4 algae R9.4: HMM F1=0.307 vs default 0.237 ($\Delta=+0.070$)

Train-test gap = 0.000 across all CV folds → robust, not overfit.

2. Chemistry-specific HMM optimum

- R9.4: STATES=4, STAY=0.93 (D2)
- R10.4.x: STATES=6, STAY=0.96 (RB_ecoli, RB_dmel, D1)

Wrong setting drops F1 by 0.1-0.5.

3. BOCD: EXP_LEN=600 universally optimal

- Default was EXP_LEN=250; CV on all 6 datasets says 600 is best.
- D1: 0.471→0.471, RB_ecoli: 0.307→0.306, RB_dmel: 0.315→0.315

4. BinSeg: speedup champion on small/medium genomes

- D1 SARS-CoV-2: F1=0.958 vs default 0.968, wall 931s vs 1359s (1.46× speedup)
- D6 R10.4 ecoli: F1=0.496 vs default 0.484 (1.11× speedup)

Per-segment penalty fix (vs global log(n) penalty) was critical.

5. Top-K saved 4 broken methods

Threshold-based detection (gradient/mad/window) gave F1≈0; switching to top-K with target_event_len=9 raised F1 to 0.1-0.5. The score function only needs to RANK boundaries roughly correctly — not produce calibrated scores.

6. RB_hsap reference-side fix (chr21-only → GRCh38)

- Default F1: 0.009 (chr21-only) → 0.210 (GRCh38, 23×)

Most reads weren't on chr21 → no truth → F1≈0. Full genome resolves the issue.

7. Negative results worth noting

- CUSUM remains weak in any reformulation (≤ 0.13) — temporal accumulation orthogonal to per-position k-mer-boundary detection.
- PELT and default produce identical events on every dataset (custom-C PELT is redundant).
- 3 attempted novel segmenters (wavelet, TDA, convbank) all underperformed default.

TDA's 1D persistence is uninformative for this signal type.

- Hand-tuned 'turbo' and 'fusion' (single-pass default) was 13% slower than default — default is already compiler-vectorised + cache-resident.

default vs pelt — same F1, pelt is faster

Dataset	default wall	pelt wall	speedup	F1 default	F1 pelt
D1	1359.1s	1350.9s	1.01×	0.9682	0.9056
D2	549.1s	333.6s	1.65×	0.4097	0.4923
D3	5.8s	2.5s	2.38×	0.2951	0.4558
D4	13.0s	11.5s	1.13×	0.2366	0.3091
D6	1331.8s	1336.5s	1.00×	0.4845	0.4399
RB_dmel	366.1s	350.0s	1.05×	0.4799	0.4913
RB_ecoli	54.6s	37.3s	1.46×	0.4188	0.4292

Why is pelt faster than default while producing IDENTICAL mappings?

- default: dual-window t-statistic at every signal sample ($O(n \cdot W)$).
- pelt: DP on SSR cost with PELT pruning ($O(n \log n)$).
- Both find boundaries that differ by +/- a few samples.
RawHash2 hashing + chained mapping is robust to that jitter:
diff on PAFs shows IDENTICAL target coordinates per read,
only the mapping-time metadata (mt:f) differs.
- Practical takeaway: pelt is a free speedup over default for any dataset where the segmenter is the bottleneck.

Key insights — Setup A (ovl $\geq$ 0.1)

1. Setup A inflates segmenter-choice importance

Under the overlap-fraction metric (TP iff query/truth overlap $\geq$ 10% of truth span), HMM and other 'narrow-span' segmenters appear to dominate small/specific datasets. This is partly an artifact of the metric: short query spans with high overlap-of-truth percentage are rewarded over wide-span methods. See PDF v2 for the distance-based comparison that matches published numbers.

2. HMM with chemistry-aware (STATES, STAY) is the Setup A winner

CV-tuned HMM consistently outperforms default on D2/D3 ovl $\geq$ 0.1 evaluation.

Chemistry-specific best:

R9.4 -> STATES=4-6, STAY=0.93

R10.4+ -> STATES=6, STAY=0.96

Train-test gap $\approx$ 0 on all CV datasets — robust, no overfitting.

3. PELT had a silent fall-through bug (now fixed)

Initial PELT runs produced bit-identical F1 to default — the result of a candidate-pruning bug at revert.c:415 dropping every candidate before it could survive to the next DP step. After the fix and per-dataset tuning (pen_mult=0.05, min_size=3), PELT reaches default F1 $\pm$ 1.5% but is 4-11 $\times$ slower than default's t-statistic. The earlier 'PELT speedup' was timing noise from the broken implementation falling through to default.

4. BOCD with EXP_LEN=600 is universally optimal

Cross-validated on all 6 CV datasets. F1 within 5-10% of default.

Useful when probabilistic interpretability matters.

5. CUSUM dropped — algorithmic dead end on nanopore

F1 < 0.1 every metric, every dataset. Page-1954 CUSUM is incompatible with nanopore translocation signal characteristics.

6. Top-K selection rescues threshold-based methods

Replace `score > T` with rank-based selection (K=signal_length/9). Boosts gradient/mad/window from F1 $\approx$ 0 to F1=0.1-0.5 on small genomes. Does not rescue them on R10.4 or large eukaryotic genomes.

7. Per-segmenter index is mandatory

Each segmenter builds its OWN RawHash2 index by simulating reference signal with the same segmenter used at query time. Cross-segmenter index reuse drops F1 to $\approx$ 0 — engineering finding, not a metric artifact.